Project Work

Instructions for project work

1. Project work may be done in small groups.
2. Each child has to do three projects - one each in chemistry, physics and biology.
3. Project writing should be done in correct order. Diagrams, pictures, paper, cuttings, collections, photographs, graphs etc. should be described as required.
4. During the practical exams, it is necessary to question each child orally regarding their projects and experiments.
5. Project work can focus on a local issue.

Biology

1. Understanding the process of natural selection.
2. Understanding the living and non-living components in our environment and also the linkages between them.
3. Understanding the role of chance in heredity.

Chemistry

1. Find the pH values of aqueous solutions of different substances using pH paper.
2. Collecting information about iron ores and methods of extraction of iron in Chhattisgarh. Linking this with metal art in Chhattisgarh.
3. Collecting information about polythene usage from families in your neighbourhood and then relating this to various efforts being made to stop use of polythene.
4. Study saponification process during soap formation.

Physics

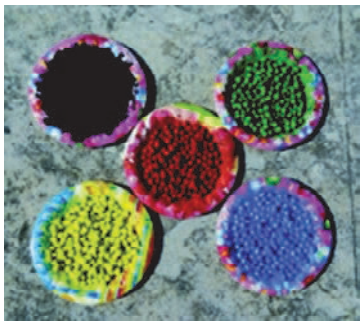
1. Making an optical instrument.
2. Producing a magnetic field by opening/closing electric circuits rapidly and continuously. By varying the intensity of magnetic field rapidly electricity can be generated.
3. Making a windmill model and demonstrating the use of wind energy in daily life.

Biology

Project work (Evolution)

Aim: To understand the process of natural selection

Materials Required: Gram (chick pea) seeds coloured separately in 5 different colours(see figure, in place of gram one may also use similar buttons of 5 different colours). Hundred gram seeds of each colour must be present. Thus a total of 500 seeds must be present. In a class of around 5 plates, string or rope and a piece of chalk is required.



Method: Allow a student to give guidelines to others to make groups each of 6 students. Two members of each group will be leaders while the rest will be birds. The students representing birds would stand around square fields. Each group should be given 12 seeds of each colour, that is a total of 60 seeds. The gram seeds represent beetles. Then each group shall be given a plate. As the game starts the 60 gram seeds would be thrown in a 1m x 1m area from where the seeds would be picked and collected in the plates.

1. Each group will mark plot of 1m by 1m area on the ground, the area marked by a group should be around 2-3 meters away from that marked by another group.
2. Try to choose such areas where the surface has grasses in some places while pebbles and stones in others.
3. The group leader of each group shall note observations in a table. Take count of gram seeds in the field and those left in the field after a round of the game. The group leaders should have a copy of the table as provided in this section(they may draw it in their copies).
4. The group leader will take a count of gram seeds in the beginning of the game and ensure that there are 12 gram seeds of each colour and a total of 60 seeds(this number has to be filled in the table in the first row). The leader shall throw the seeds over the plot as four members of the group shall stand around four sides of the plot. These represent birds. Their back should be towards the plot so that they are unaware of the distribution of the gram seeds.
5. Each of the empty plates should be kept at a distance of at least around 3-4 meters from the respective plots.
6. One student,not a member of any group, would have to signal the start of the game. The students representing birds should immediately turn collect a gram

seed and run towards the plate, put it on the plate and return to collect the next seed. After around 6 rounds the same student who signaled the start of the game will signal the stop of the game.

7. Now the group leader present near each of the plates of the respective groups should count the seeds of each colour and find out the number of seeds of each colour left on the field. They should write the numbers in the given table.
8. The group leader would now add double the number of each colour of seeds left on the field(doubling the number just represents the process of reproduction of beetles with a progeny population double as that of parents). The leader would have to sprinkle them over the respective fields. Those that are representing birds should be standing with their back towards the field so that they do not see the process(Steps 1-8 represents 1st round).
9. Repeat processes 4 to 8(this would be second round). That is we progress up to two generation of beetle population after parental generation.
10. Now students may return to their class, complete the table, analyze data obtained in the table, draw graphs and find out what had happened to the beetle population.

S.No.	**No. of seeds (representing beetles)**	**Colour**				
		Black	**Blue**	**Green**	**Yellow**	**Red**
1.	Parental generation(No. of seeds at the start of experiment)					
2.	No. of seeds left on the plate after first round					
3.	No. of seeds left on the field					
4.	No. after doubling values obtained in '3'					
5.	Total number on field (add 3 and 4.					
6.	No. of seeds on plate after second round					
7.	No. left on field					
8.	Double number left on field					
9.	Total seeds on field					

Use the numbers obtained in the last row(row 9) for graphical representation.

1. Did the relative number of each colour change a lot after the first round?
2. What happened after the second round?
3. Do you think that in both the generation the choice of selection of seeds was same?
4. Which colour of seeds were most in the fields?

5. Which colour of seeds was least or absent?
6. What would happen if you continued with your game up to 3rd ,4th or further generations?
7. Continue playing to check your hypothesis.
8. What was your experience after playing the game?
9. What do you think may be the reason for your observation?

Consolidation: Discuss the process of natural selection as observed in the game and differentiate between artificial selection and natural selection. State that natural selection is a non-intentional process, continuously going on in nature.

Project work (Our Environment -Energy flow in ecosystems)

Aim: To use transact method to find out interrelationships in ecosystem

Materials required: four wooden sticks to mark four corners of 1m X 1m square area. A length of strong thread to outline the chosen area, a spade or hoe to dig soil, copy, pencil, rubber, sharpener , hand-lens

Procedure: Select an area around your house or school. It could be a part of a garden or a grassy field. The area should be such that you could observe it regularly (at least twice daily, both in the morning and evening) for 5 days. Mark a one meter by one meter area here. Observe carefully, do you find any insects, frog, bird or any other organism in the marked area? Write the name of the group of insects and the number of them that you observe there (as for example grass 20, ants 28 etc.). Dig the soil as well to count organisms. Try to keep counts such that you count a particular organism once only. If there are any trees, herbs, bushes etc. count them under the group of plants.

Note your observations every day. Write the date, morning and evening half and details to be able to take counts daily.

Questions:

- What are the biotic and abiotic factors in your area?
- What are the biotic and abiotic factors in your area?
- Could you observe any food chain or food web in the marked area? Give an example.
- Could you observe any food web or food chain in the whole garden? Give an example.
- What are the differences in abiotic factors of morning and evening hours?

Now calculate the average number of organisms observed per day in the area from the total organisms observed in 5 days(as for example if total number of ants were 50 those observed per day were 10 and this number multiplied by the area gives you and approximate number of ant population in that area).

- Do you think the organisms in the area have enough resources for their survival?
- What do you think will happen if there aren't enough resources for survival? Elaborate your answer.

Projcet work (Life Processes: Growth, development and reproduction)

Aim: Study development and rate of growth in mustard plants

Materials required: Copy, scale, a length of thread(around 1m), pencil, pen, sharpener, rubber, hand-lens, a medium sized pot filled with soil, mustard seeds, cup.

Procedure: Take a few mustard seeds and sow them in a medium sized pot. Note this day as day 1.Water the pot every day with around a cup of water. Observe the pot and see when the plants grow out of the soil. Note the day. Keep observing till the plants are around 30cm in length.

Note your observations carefully.

1. How many days after sowing the seed did you find the plants growing out?
2. When did you find a pair of leaves growing on the plants?
3. When did the next pair of leaves grow?
4. Was there any change in the first pair of leaves as the plant grew taller?
5. In how many days did the plants grow to nearly 30 cm in height?
6. Find out the approximate rate of growth of mustard plants(Hint: divide plant length by number of days after sowing).

Projcet work (Heredity: Parent to offspring)

Aim: Tossing coins to understand probability and how it may be applied to the process of heredity.

Materials required: two similar coins either of one rupee/two rupees/ 5 rupees, marker pen(preferably black), paper, pencil

Instruction: Draw table 3 as given in chapter 15 on paper.

Procedure:

1. Start the game of tossing coins by using two similar coins(one rupee/two rupees/5 rupees). Mark one of the coins on both sides with a dot using the black marker pen. Mark it in such a manner that the head and tail marks are not covered. This is coin 'A'. The one without marks is coin 'B'.
2. Now toss both coins together and note your observation in the table with tally marks for example if both fall with head side up then one tally marks to be put under section HH, with a head on 'A' coin and tail on 'B' put tally mark under HT, tail on 'A' and head on 'B' put a tally mark under TH and so on.

3. Toss coins at least 500 to 1000 times
4. Find out the percentage probability of getting two heads; two tails; head on A and tail on B; tail on A and head on B.
5. Consider HH and TT as alternative traits of a character(say HH represents tallness and TT represents dwarfness).
6. Take the help of Chapter 15 especially section 15.2.4.

Questions:

1. Suggest how Mendel may have used probability to predict inheritance of characters from one generation to the next?
2. What do you think would happen if a single coin was tossed? What is the probability of getting a head? Why do you think Mendel may have suggested that characters are passed on from one generation to next as at least a pair of traits?
3. How do you think mathematical calculation of probability may have helped Mendel to arrive at his proposed laws of inheritance?

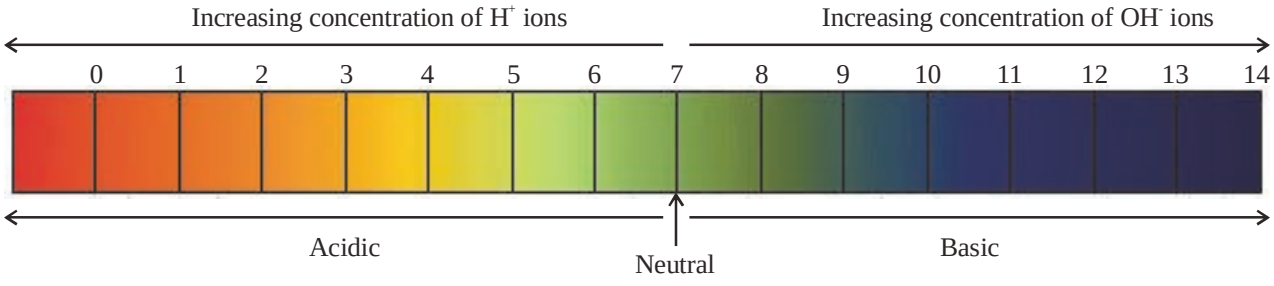
Chemistry

Project work (Acids, Bases and Salts)

Objective **:** Find the pH values of aqueous solutions of different substances using pH paper.

Essential Apparatus: Different test solutions as follows: 1) solution of soil in water 2) fruit juice 3) milk 4) solutions of different substances, test tubes, test tube stand, pH paper, glass rod or droppers.

Theory **:** pH is a measure of the acidic or basic nature of a dilute solution. pH scale tells us the concentration of H+ ions in a given solution. pH is divided on a scale of 0 to 14. At 25°C (298K), the pH of a neutral solution is 7. Value of less than 7 on the pH scale shows acidic nature and that of more than 7 shows basic nature. Usually, pH paper is used to find the approximate pH value of any solution. It indicates different pH values through different colours.



Procedure : (Note- all solutions should be made in distilled water. A dropper or glass rod used with any one solution should be thoroughly cleaned before with water before using again.)

1. Take clean test tubes in a test tube stand.
2. To make solutions of solid substances, add a pinch to about 10 mL of water. Use liquid substances like fruit juice as they are.
3. Take different strips of pH paper. Using a dropper or glass rod, put one or two drops of the different solutions on different strips of pH paper.
4. Match the colour obtained with the pH paper chart to determine the pH of each solution.
5. To find the pH of fruit juice, squeeze out a few drops on pH paper and note the pH.

Observation Table :

S. No.	test solution	approximate pH value
1.	solution of soil	
2.	fruit juice	
3.	milk	
4.		
5.		
6.		
7.		
8.		
9.		
10.		

Conclusions : Identify the nature (acidic, basic, neutral) of different solutions and write the conclusions.

Project work (Metals and metallurgy)

Objective : Collecting information about iron ores and methods of extraction of iron in Chhattisgarh. Linking this with metal art in Chhattisgarh.

Theory : Ores are found in large quantities in Chhattisgarh and our state holds an important place in the country in terms of iron production. The ores are found in different types of rocks. Ores are also formed due to soil compaction and other factors. Through this project, we will be able to understand that iron ore-

1. Is found in which types of rocks in Chhattisgarh.

2. What are the methods of extraction used?
3. How is the iron used in the field of metal art?

Procedure **:** 1. Groups of students should be formed by the teacher.

2. During project work, each group should collect information from their surroundings around the following questions:

- Where in Chhattisgarh is iron ore found?
- What were the local methods of iron extraction used previously and what are the current methods?
- What is the history of iron metal art in Chhattisgarh?
- What is the chemistry behind metal art?
- What are the traditional alloys of Chhattisgarh?
- Make a list of objects made of different alloys found in and around your home or school.
- How to different cultures connect to metal art and how is the art connected to local context?

Prepare the project report according the answers given to your questions.

Project work (Derivatives of hydrocarbon)

Objective **:** Collecting information about polythene usage from families in your neighbourhood and then relating this to various efforts being made to stop use of polythene.

Theory **:** People continue to use objects made of polythene even today. What was used instead of polythene 40-50 years ago before its invention? Is the decomposition and recycling process for paper and polythene similar?

Since polythene is inert to most chemicals therefore it keeps collecting as waste and does not decompose. In this situation, the polythene dumped in garbage is either burnt to destroy it or recycled to make secondary products. Both these processes effect our environment. Through this project, reflect on the following questions:

1. Why has polythene been banned in our state and country?
2. What efforts are being made to stop the use of polythene?
3. What products are formed during combustion of polythene and how do they effect the environment?
4. Which objects are formed by recycling of polythene?

Procedure **:** 1. Groups of students should be formed by the teacher.

2. During project work, each group should make a questionnaire to interview members of four families in their locality. After the interview, prepare a project report.

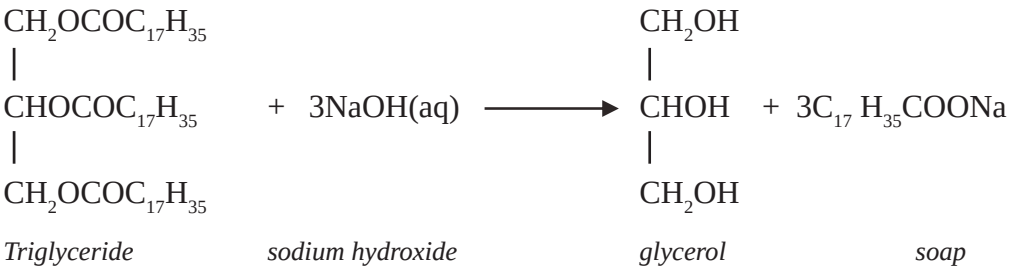
3. If articles from newspaper or magazines have been used or any other material has been used during report preparation, its list should be prepared by each group.
4. If new information or new questions came up during project work they should be shared with the teacher and other groups and included in the report.
5. Also include any efforts made by student groups to increase community awareness on the issue.
6. ach group can prepare posters, small skits, songs etc. in an effort to increase community awareness on the issue.

Project work (Chemistry in daily life)

Objective **:** Study saponification process during soap manufacture.

Necessary material and chemicals: sodium hydroxide, vegetable oil (for example 20 mL castor oil), 10 g common salt, distilled water, red and blue litmus papers, two 250 mL beakers, two test tubes, a glass rod, 50 mL measuring cylinder and a knife.

Theory **:** when an oil or fat is reacted with sodium hydroxide then a sodium salt of the acidic oil (soap) and glycerol are formed. This is known as saponification.



This is an exothermic reaction which means that heat is released during formation of soap.

Procedure **:**
1. Take 20 mL castor oil in a 250 mL beaker.
2. In distilled water, prepare a 20 % solution of sodium hydroxide (10g NaOH in 40 g of water) and add 20 mL of this solution to the castor oil.
3. Dip red and blue litmus paper in the reaction mixture to test its nature. Did you notice any change in colour of any of the litmus papers. Note your observations.
4. Touch the external beaker wall - is it hot or cold?
5. Add 5 - 10 g common salt to the mixture and continuously stir the mixture with a glass rod till soap starts forming.

6. Leave the reaction mixture undisturbed for a day so that the soap solidifies and hardens.
7. Remove the soap cake and cut into desired shape and size.
8. If castor oil is not available, use any other vegetable oil.
9. Scent and colour can also be added to the soap.

Observations **:**
1. On dipping in the reaction mixture, the colour of red litmus paper is …….. and that of blue litmus paper is ……
2. The temperature of reaction mixture ……(rises/ falls) on adding sodium hydroxide to the oil.

Conclusions **:** On the basis of your observations of litmus paper, decide whether soap is acidic or alkaline. Also tell whether saponification is exothermic or endothermic.

Glycerol is formed as a second product during saponification. Soap is salt of oil acid and is its precipitate.

Precautions **:**
1. Stir the reaction mixture carefully so that it does not spill out of the beaker.

Physics

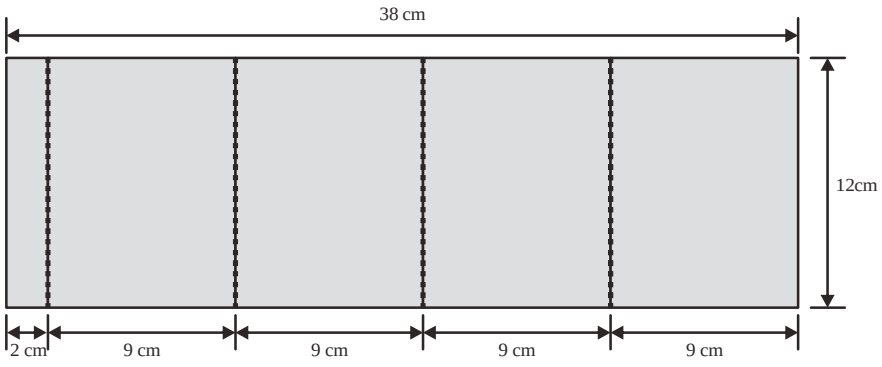
Project Work (Light: Reflection and refraction)

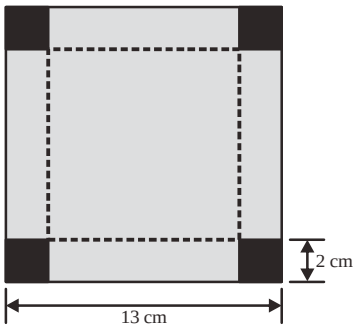
Objective **:** Making an optical instrument.

Theory **:** Pinhole camera, microscope and telescopes also are based on the principle that "light travels in straight line".

Procedure **:** You need stiff black paper to make two special types of boxes.

For the first box, cut a 38 cm long and 12 cm wide strip of black paper. As shown in the figure, fold the strip at four places at a distance of 9 cm from each other. You will be left with a 2cm wide strip at one end. Put glue on it and stick it to the opposite end. You now have a window box where both sides are open.

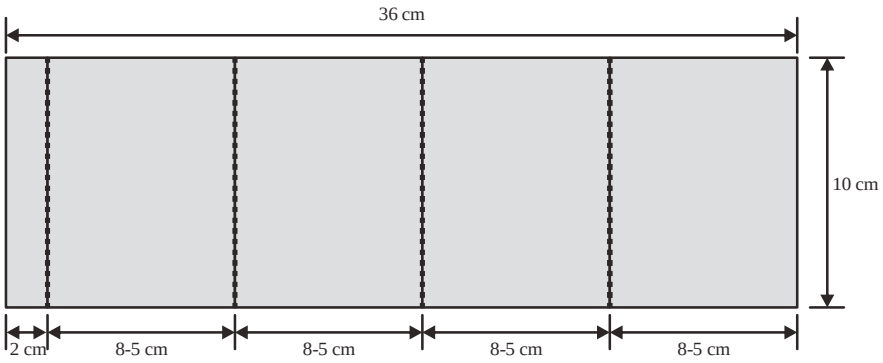




Now, cut a square of 13 cm side from the black paper. Cut a 2 cm square from each of its sides as shown in the figure. Fold along the dotted line and stick this to one end of the first box to form its base.

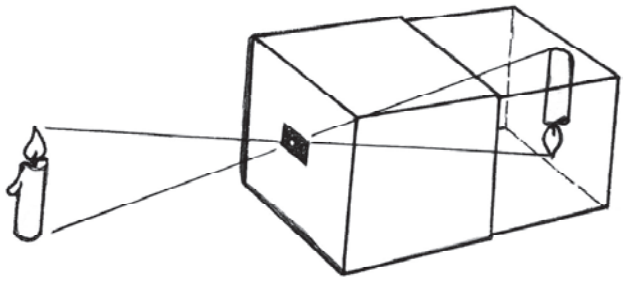
Your first box is ready. Make a small hole on its base using an awl-pin.

For the second box, cut a 36 cm long and 10 cm wide strip of black paper. As shown in the figure, fold the strip at four places at a distance of 8.5 cm from each other. Now glue it to the opposite end, as described for the first box. Stick white paper carefully on one end of this box. Put an oil-drop on the white paper to make it slightly semi-transparent. This is your screen box.



Make your own pinhole camera

Take the two boxes. Put the curtain box inside the window box - your pinhole camera is ready. Put a lighted candle in front of the hole of the camera and see from the other end.



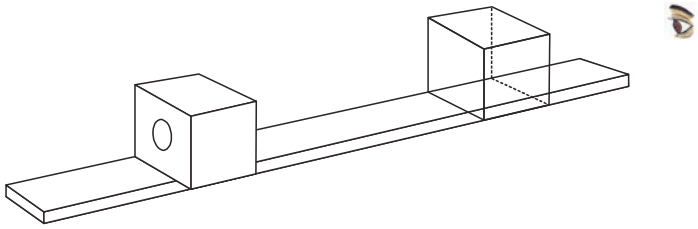
What do you see on the screen? Move the screen box forwards and backwards and note what happens to the candle image each time. Does shifting the screen cause any change in the image?

How do you think this image was formed? Look at the picture and think of the answer.

Through the camera, look at an object which has a lot of light falling on it, for example, a tree or a building. Now look at your camera screen. If light is falling on the camera, cover it with your hands.

Make your own binoculars

To make binoculars you'll need to fix a thick convex lens as well as a think convex lens to the box. To do this, cut a 3



cm circular hold in the screen and stick the thin lens to it. Keep the thick lens near your eye and the thin lens 40 cm away from it. Move the thin lens forwards and backwards. At one point, a far-off object will appear close. Look at far-off objects (houses, trees etc.) with your instrument; how do they appear?

Making a microscope

Take a fat lens in your hand and look at the letters in your textbook. First the letters appear very big but when you start to move the lens away they become smaller and then appear inverted.

Keeping the thick lens in the final position, place the thin lens in front of your eyes and look at the letters. You can lower the lenses to make the image clearer.

Do the letters look big?

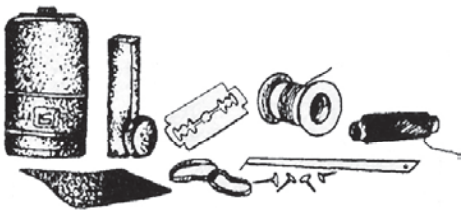
How is the image, right side up or inverted?

Project work (Magnetic effects of current)

Objective **:** Producing a magnetic field by opening/closing electric circuits rapidly and continuously. By varying the intensity of magnetic field rapidly electricity can be generated.

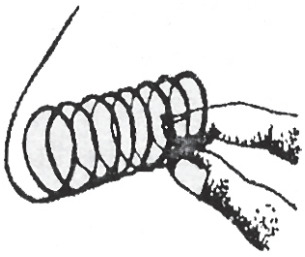
Theory **:** Different objects around us like fans, tape records, electric bells, etc. function when a magnetic field is produced in an electric field.

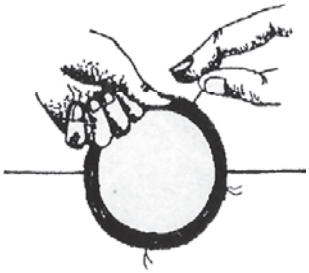
Necessary Apparatus: 1. 24 gauge copper wire used in motor rewinding, about 1 m long. This is usually available in electrical shops. This does not have plastic cover, only an enamel coating.



2. Torch cell
3. Circular magnet or bar magnet.
4. Blade or sand paper
5. Stove pin
6. Cycle tube cut into bands
7. Nail
8. String

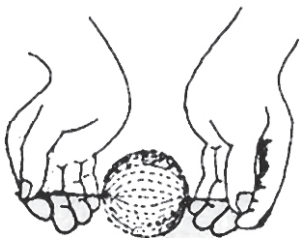
Procedure **:** First coil the wire tightly around the cell to form a coil.





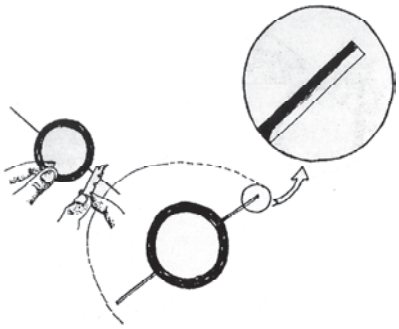
To ensure that the coil does not open when it removed from around the cell, tie it with string at a few places. Or you use tape or bend the open ends.

Since this coil is the most important aspect of the experiment, make it very carefully. Both ends of the coil should line on the lie passing through the center of the coil.

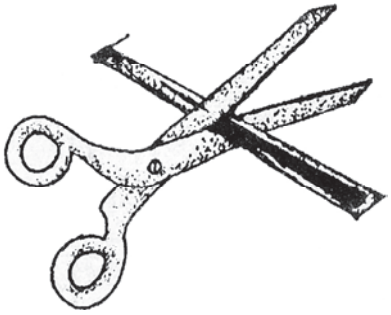


If the coil is properly aligned on this axis, it will move freely and smoothly. To test the balance, you can use your fingers and turn it as shown in the figure. If it moves easily and for a long time then it is ready. If for some reason, weight is more on one side you will immediately see that the coil is not spinning properly and jerks and stops.

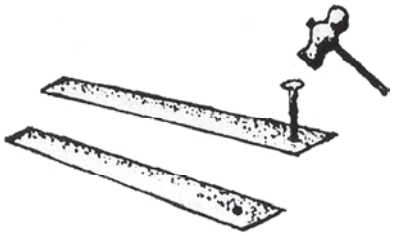




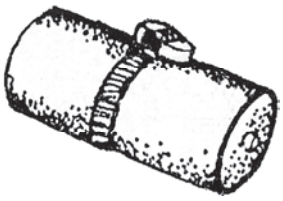
Now you will have to sand away the enamel from the upper half of each end of the coil. Do it carefully and do not remove the paint completely. Rub only from one side as shown in figure. If you remove the paint completely, the motor will not run. You should have copper on half of the wire and paint on other half.



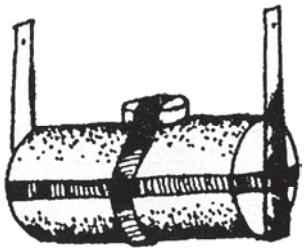
In this way, by keeping half of the part insulated and other conducting, we have prepared a method to close and open the circuit quickly, as was mentioned in the objective.



Once your coil is ready, cut the stove pin into two equal parts. Using a nail, make small holes on the ends of both the pieces.



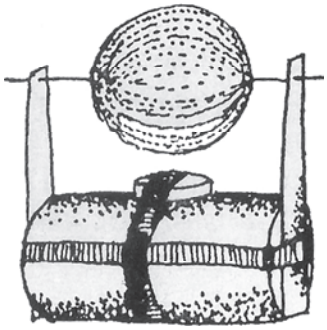
If you are able to find circular magnets, attach them to the cell with the cycle tube bands. Actually, using a bar magnet or circular magnet we have to ensure that only one pole is towards the coil.



Put another cycle tube band around the cell. Use this to fix the stove pins to the two ends of the cell, the positive end and the negative end. They should be touching because only them will the circuit be complete.

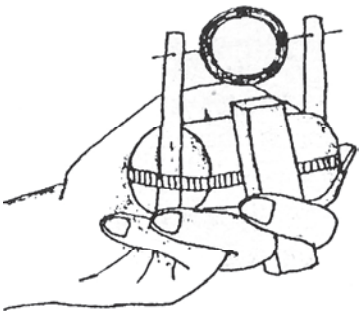


Spread the pins a little and fix the coil in the holes in them.



Check each part of the circuit carefully and gently push the coil. Did it start spinning?

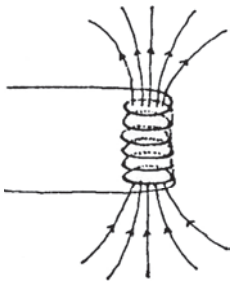
If the coil stops spinning, push it in the other direction.

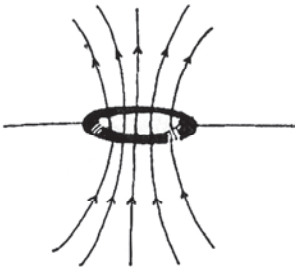


If the motor does not work even now, then you will have to recheck the following - balance of coil, have the ends been correctly sanded or has the enamel been completely removed, connections are correct, is there any rust, is the coil very far from the magnet etc. If the motor starts working properly we can do many activities with it.

- You can fix a picture to the coil to understand persistence of vision.
- You can see what happens if we change radius of coil, number of turns, shape of coil etc. How do they affect the rate of spinning and its direction?
- If you turn over the motor, you will observe something interesting.

Once the motor is made, let us try to understand why the coil spins and how the motor works.



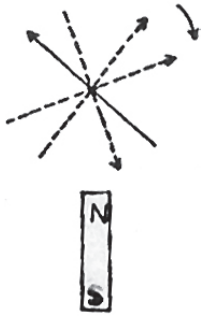


When current passes through any coil then it forms a magnetic field whose effect is similar to that of a bar magnet.

This means that whenever the clean part of the coil i.e. the conductor comes in contact with the stove pin then electric current passes through the coil and it starts behaving like a magnet.

This means that we have a still, flat magnet on the bottom and a freely moving bar magnet (in this case, the coil) over it which on spinning behave half the time like a magnet and like a rod the other half of the time.

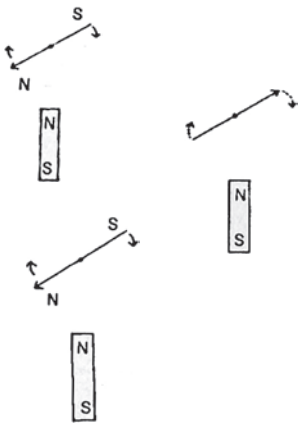
When the clean part of the coil comes in contact with the stove pin, then electric current flows and the coil behaves like a magnet. The other half of the time, when enamel part is in contact with the stove pins there is no flow of current and the coil is not a magnet.



What happens when we gently push the coil and set it in motion? Let us understand through some diagrams.

Suppose the clean part of the coil comes in contact with the stove pin, then electric current flows and the coil behaves like a magnet. Then the like pole of the magnet below the coil will be repelled and pushed and the opposite poles will be attracted towards each other and the coil will spin.

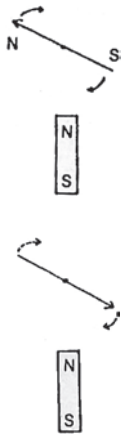
This is the same situation, N is repelled and S is attracted.



Once half-spin is complete, then enamel comes in contact with the stove pin and electric current stops flowing. But because of the earlier motion, the coil continues to spin.

Although the current is not flowing yet the soil is spinning due to inertia because of which the conducting portion again comes in contact with the stove pin, circuit closes and current flows again pushing the coil.

This process continues, the coil spins and this is our motor.



Project work (Energy: Sources and Forms)

Objective : Making a windmill model and demonstrating the use of wind energy in daily life.

Material : tin sheet, tin cutter, cycle spoke, stand, pump to remove kerosene oil, scale, protactor, water, container etc.

Principle : Flowing air is called wind. Due to kinetic energy, wind is capable to work. When wind comes in contact with windmill blades, a force is applied on them which causes the wheel to rotate. By using the constant motion of the rotating fans, wind energy can be produced.

Procedure :

1. Cut a big circle from the tin sheet using tin cutter.
2. Divide the perimeter into six equal parts $\left(\frac{360^\circ}{6} = 60^\circ\right)$ and cut each part a little way from the perimeter to the center (do not cut completely).
3. Bend and fold one side of each part as shown in the figure. This is the wheel of the windmill.

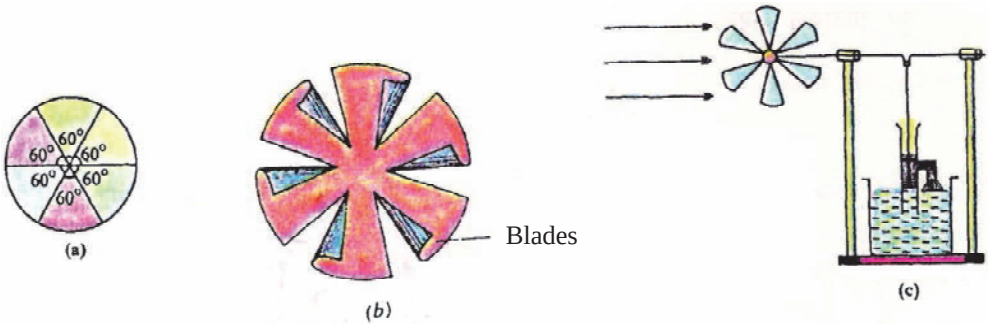
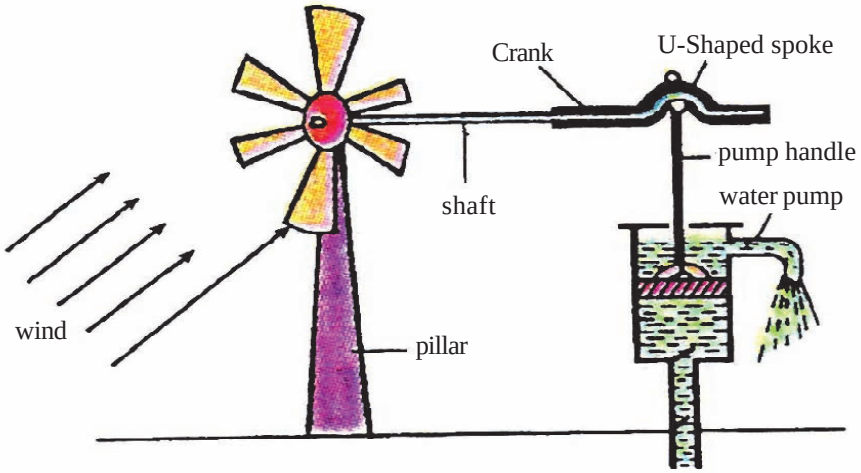


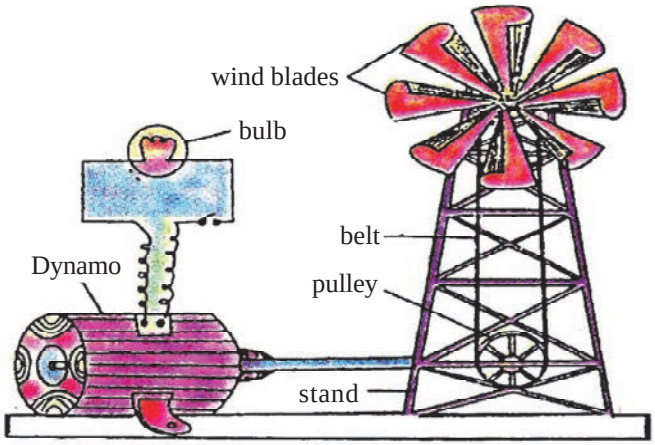
Figure - water pumped up be a windmill

4. Now bend the cycle spoke into U shape from the centre as shown in the figure.
5. Attach the pump handle to this U-shaped spoke.
6. Fix the cycle spoke into a stand as shown in the figure.
7. On one end of the spoke attach the windmill in such a manner that the spoke also moves along with the windmill.
8. Dip the pump into a water container containing water so that the water removal part is outside.
9. Now place the windmill in front of a wind source so that the wheel is in front of the source.

Observations : When the blades of the windmill rotate in the wind, the windmill spins moving the spoke as well as the pump handle with it. This pumps out water. This set-up is used to draw water from wells and also underground water (figure).



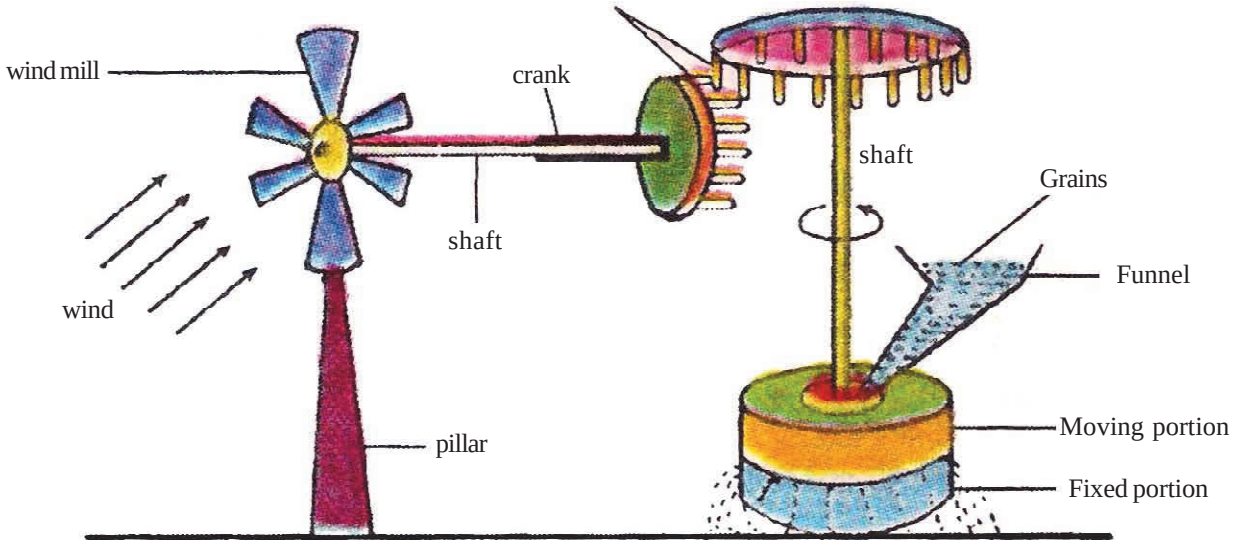
Tips : 1. If there is no wind, then use a table fan as wind source. Windmill does not work in low speed wind conditions.



2. If a dynamo is attached with the spoke using a belt (as per the figure) then even electricity can be produced.

   Similarly, windmills are used for grinding wheat.

Results : This model of the windmill proves that the kinetic energy of wind can be converted into different other forms of energy and utilized in our daily life.



Precautions :

1. The blades of the windmill should be in the same direction as the direction of wind.
2. Pump should be light.
3. The fans of the windmills should be light and hollow.
4. Windmills should be placed in areas where there the wind flows at high speeds.

Benefits :

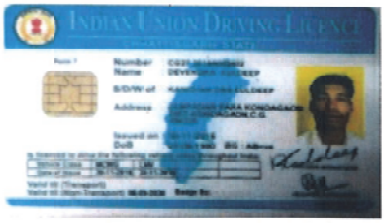
1. It conserves conventional energy sources like coal and oil.
2. It may help us overcome energy crisis in the future.
3. There is no environment pollution.
4. The energy obtained can be transformed into other forms.
5. On large scale, it can be used to produce electrical energy as has been done in Gujarat state.

Limitations : 1. We cannot use it continuously.

2. It is not effective in areas where the speed of wind is low.
3. It can convert 60-70 % of wind energy into work.
4. If winds are not available, it can't be used.
5. To set up windmills requires large areas.

-: वाहन चलाते समय वाहन में रखे जाने वाले महत्वपूर्ण दस्तावेज :-

चालक अनुज्ञा पत्र :- वाहन चलाते समय चालक को वैध अनुज्ञा पत्र (ड्रायविंग लायसेंस) साथ रखना अनिवार्य होता है।



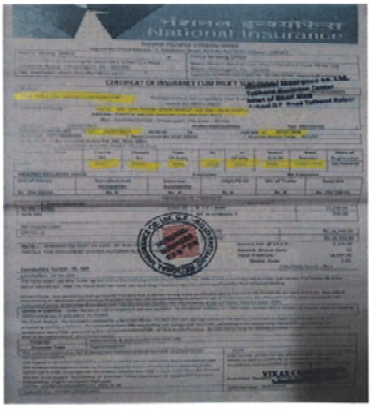
पंजीयन प्रमाण- पत्र :- सभी प्रकार के डीजल/पेट्रोल/इलेक्ट्रिक पावर से चलने वाले वाहनों के लिए परिवहन विभाग द्वारा जारी पंजोयन प्रमाण-पत्र (रजिस्ट्रेशन सर्टिफिकेट) रखना आवश्यक होता है।



फिटनेस प्रमाण-पत्र :- व्यवसायिक प्रयोग में आने वाले वाहनों के सभी कलपुर्जों को दुरूस्त रखना अतिआवश्यक है। अतः ऐसे वाहनों को फिटनेस प्रमाण-पत्र की आवश्यकता होती है।



बीमा प्रमाण-पत्र :- सभी प्रकार के डीजल/पेट्रोल/इलेक्ट्रिक पावर से चलने वाले वाहनों के लिए बीमा प्रमाण-पत्र (इंश्योरेंस सर्टिफिकेट) रखना आवश्यक होता है।



प्रदूषण प्रमाण-पत्र :- डीजल/पेट्रोल चलित वाहनों के लिए कार्बन उत्सर्जन को नियंत्रित रखने तथा पर्यावरण को बनाये रखने हेतु पदूषण नियंत्रण प्रमाण-पत्र रखना आवश्यक होता है।

